

Dual-adaptive interactive transformer with textual and visual context for image captioning

Lizhi Chen^a, Kesen Li^{b,*}

^a Soochow University, School of Software, Suzhou 215000, China

^b Jiyang College of Zhejiang A&F University, College of Engineering Technology, Zhuji 311899, China

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ABSTRACT

The multimodal Transformer, which integrates visual and textual contextual information, has recently shown success in image captioning tasks. However, there is still natural complementarity and redundancy between text and vision, and effectively integrating the information from both modalities is crucial for comprehending the content of an image. In this paper, we propose the Dual-Adaptive Interactive Transformer (DAIT), which incorporates similar textual and visual contextual information into the encoding and decoding stages. Specifically, during encoding, we propose the Adaptive Interactive Encoder (AIE), which expands the feature vectors for both modalities through the introduction of new operations. Furthermore, we also introduce normalization gate factors to mitigate noise caused by the interaction between the two modalities. During decoding, we propose the Adaptive Interactive Decoder (AID), which adaptively adjusts the multimodal features at each moment through similarity-weighted textual and visual branches. To validate our model, we conducted extensive experiments on the MS COCO image captioning dataset and achieved outstanding performance compared to many state-of-the-art methods.

1. Introduction

Image captioning models are typically developed through an encoder-decoder architecture (Vinyals et al., 2015; Wang et al., 2021; Wang et al., 2022) which leverages Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for feature extraction and sentence generation, respectively. In recent times, the Transformer architecture (Huang et al., 2019) has further been applied to visual language tasks with great success (Kenton and Toutanova, 2019; Brown et al., 2020). Its self-attention mechanism is particularly noteworthy as it enables the capturing of correlations between features both within and between modalities. Incorporating self-attention modules can lead to improved performance and generation quality of image captioning models by more effectively modeling the relationship between text and image features.

Early image captioning research initially concentrated on processing image content information (Anderson et al., 2018; Herdade et al., 2019; Guo et al., 2020). These methods heavily relied on pre-trained features and fixed object detectors, allowing models to leverage the abundant information available in large-scale data and prior visual understanding. While this single modality approach is effective, a major limitation is

that the model's output depends solely on the results of the object detector, and modern image captioning frequently involves multiple interdependent modalities, rendering these approaches unresponsive to modality correlations. Recently, many works have combined text and visual modalities for image captioning tasks (Kuo and Kira, 2022; Ramos et al., 2023). However, text features and image features exhibit differences in information expression and perception. Text features often contain more semantic information. Due to its expressivity, text can convey a plethora of conceptual, relational, and logical information, thereby permitting more detailed and semantic information about the text content to be obtained. Conversely, image features tend to focus more on spatial and regional detail information of the image. Image is usually in pixels and has rich visual attributes such as color, texture, and shape. However, compared to text, images have a lower expressive capability for abstract concept logic reasoning.

The exploration of handling visual and textual information modalities is an important area that has seen significant advancement in recent times. However, there are still two primary challenges that need addressing. Firstly, while numerous works have proposed new fusion methods (Luo et al., 2021; Ramos et al., 2023), these typically involve simple concatenation between different feature types, subsequently

* Corresponding author.

E-mail addresses: 20234027010@stu.suda.edu.cn (L. Chen), 2019210516034@stu.cqnu.edu.cn (K. Li).

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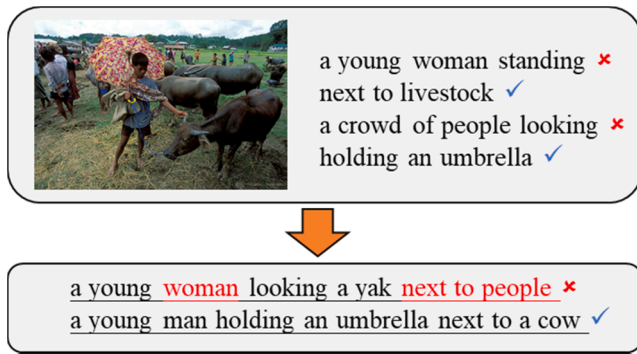


Fig. 1. An example of generated auxiliary text descriptions. in the relevant text, × indicates irrelevant information and ✓ is used for additional context. In the generated descriptions, any phrasing in red indicates an error in describing the image.

leading to detailed information loss. Secondly, many works overly rely on multi-modal fusion features (Wang et al., 2022; Kuo and Kira, 2023), resulting in information redundancy and disregarding irrelevant textual information. Specifically, in text-assisted image captioning, each image may have several associated textual information, but some of the text may not be relevant to the image and may even introduce target noise, leading to an inaccurate description. For instance, in Fig. 1, some semantic information like “a young woman” and “people looking” may not be entirely relevant to the image, introducing noise from redundant or unrelated objects.

In order to address the issues, we propose the Dual-Adaptive Interactive Transformer (DAIT), which consists of the Adaptive Interactive Encoder (AIE) and the Adaptive Interactive Decoder (AID). In the AIE, we designed the Interactive Fusion Module (IFM). This module deconstructs feature vectors into expanded matrices and delineates two interactive operations between vector and matrix, thus exploring all conceivable interactions among different modal vectors. Additionally, to alleviate the problem of noise interference between modalities, we propose a Correlation Aware Module (CAM). It incorporates normalized gate factors in autoregressive fusion operations, identifying correlations among modes which enables the model to optimize its utilization of pertinent modal information more efficiently. In the AID, we designed the Adaptive Guidance Module (AGM), which encodes existing headlines into context vectors. By introducing the AGM, the decoder can use prior knowledge of the existing headline to guide the generation process and provide additional context and semantic information. Then, the context vector is interacted with the text and visual branches using a similarity-weighted approach. In a nutshell, the contributions of this paper can be summarized:

- We propose a dual Adaptive Interactive Transformer (DAIT) model, leveraging context information of both text and visuals, to investigate the mutual benefit and significance of these modalities in image captioning.
- We propose an adaptive interaction encoder (AIE), which employs the expansion matrix to explore all possible interactions among the modalities. Moreover, the adaptive modality information is utilized to more accurately capture critical signals and filter noise.
- We propose an Adaptive Interactive Decoder (AID), which globally calculates text and visual information at each time step, with a selective focus on the most critical segment in each modality, to augment model expressivity.
- We extensively evaluated our proposed model on the benchmark MS COCO and Flickr30K datasets. The experimental results suggest that our methodology surpasses other state-of-the-art methods.

The remaining parts of this paper are organized as follows. We

review the current representative-related works in Section 2. Section 3 describes the framework of DAIT and displays the design of each layer in detail. Experimental results and concrete discussion are illustrated in Section 4. Finally, Section 5 concludes our work and outlines directions for future work.

2. Related works

Image captioning task is a prominent and enduring topic within the realm of cross-modal domains. This task can be bifurcated into two categories depending on diverse fusion structures: Traditional image captioning methods and Transformer-based image captioning methods. Below we introduce their details separately:

2.1. Image captioning

With the development of deep learning technology, the image description methods based on deep neural networks have been widely researched and show superior performance. Existing image description methods typically follow the encoder-decoder paradigm (Yao et al., 2017; Yang et al., 2019), which first uses convolutional neural networks to encode the image and then uses recursive neural networks as decoders to generate sentences. In order to imitate the ability of humans to describe different words in different regions, an attention mechanism is introduced in this task. (Xu et al., 2015) proposed a dynamic image encoding method based on grid features, which can selectively focus on specific areas of the image while generating corresponding text. (Lu et al., 2017) introduced an adaptive attention model that combines with visual sentinels, which can adjust the objects being observed at various time periods.

To address the problem of grid features lacking specific object-level semantic information in the past, (Anderson et al., 2018) proposed a method that uses a target detector (Ren et al., 2017) to extract salient objects and region features and calculates attention through top-down approach. They trained a target detector and used attribute prediction as the loss function, which performed excellently on the Visual Genome (Krishna et al., 2017) dataset. (Chen et al., 2022) embed relationship features in global-local attention and created a new pyramid attention mechanism that explores intrinsic visual and semantic relationships between different object regions. (Zhang et al., 2021) proposed a context-adaptive attention module to enhance the performance of the decoder by enriching the language context content. (Yang et al., 2023) proposed a new image captor that learns the pairing of language neural modules. The models mentioned above directly utilize the visual features of the input image and generate descriptions using a semantic decoder. No extrinsic multi-modal semantics are incorporated into the input information, which may lead to an improper alignment between the visual features and the semantic space of the decoder.

2.2. Multi-modal feature encodings

With the success of the Transformer architecture in the field of natural language processing (Huang et al., 2019), recent work (Fei, 2021; Luo et al., 2021) has completely replaced RNNs with self-attention as the decoder, resulting in new state-of-the-art performance. In self-attention mechanisms, each element is associated with other elements. This mechanism can be used to compute the representation of the collection of features after interaction, similar to the earlier use of self-attention in image region features or semantic tag sets (Yang et al., 2019). In recent years, there has been increasing interest in self-attention for image features. (Yu et al., 2019) proposed a multimodal Transformer model that treats image features obtained from different visual feature extractors as different views. (Ji et al., 2021) introduced the global-enhanced Transformer to extract more comprehensive global representations.

Similar to our work, (Wang et al., 2022) devised an innovative

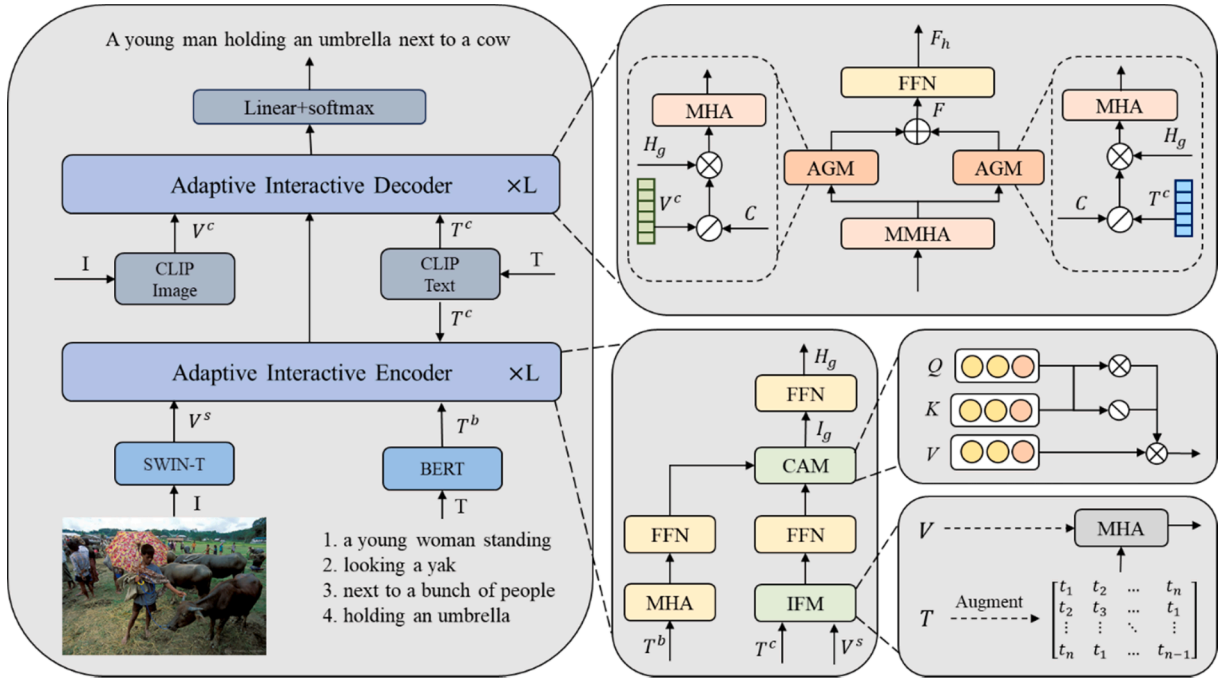


Fig. 2. The overall architecture of our DAIT.

method for embedding tree structures, which narrows down the prediction scope of words by implementing hierarchical text semantic modeling, and generates a set of hierarchically representative embeddings to capture the hierarchical semantic structure within the text space. (Kuo and Kira, 2022) proposed a method of adding auxiliary inputs to represent missing information, by mining attributes and relationships in the Visual Genome dataset to obtain missing information and introducing it into the description model. Although these methods have achieved state-of-the-art performance, the fusion of visual and textual information is achieved through a simple cascade operation, which is intuitive but lacks in-depth analysis. In addition, exploring the use of both visual and textual global information simultaneously in the decoding process to complete the image captioning task has never been attempted, which motivated our work in this paper.

3. Method

In this section, we propose the Dual-Adaptive Interactive Transformer (DAIT) model, the comprehensive structure of which is depicted in Fig. 2. Initially, we elucidate the extraction of distinct modality characteristics via the Swin-T, BERT, and CLIP models. Then, we succinctly outline the fundamental principles of a conventional Transformer architecture. Subsequently, we proceed to introduce the DAIT framework apparatus for image captioning, which comprises the Adaptive Interactive Encoder (AIE) and Adaptive Interactive Decoder (AID). In the encoding, we propose the Interactive Fusion Module (IFM) and Correlation Aware Module (CAM). During decoding, we propose the Adaptive Guidance Module (AGM).

3.1. Multi-modal feature encoder

3.1.1. Visual feature encoding via Swin-T

Swin-T (Liu et al., 2021) is a visual Transformer that achieves state-of-the-art performance on many computer vision tasks such as object detection and semantic segmentation. In this paper, we divide the input image $I \in \mathbb{R}^{H \times W \times 3}$ into B non-overlapping patch regions, with the local area denoted as $I_p \in \mathbb{R}^{p \times p \times 3}$. Here, $\{H, W\}$ and $\{P, P\}$ respectively represent the size of the input image and the patched area, and $N =$

$(H \times W)/P^2$ is the data of the patched area. To preserve the positional information, position embedding is also integrated into the patched embedding vector. We concatenate the patched embeddings of the image to obtain the visual sequence of patched embeddings $V_{pc} \in \mathbb{R}^{N \times d_v}$.

$$V_0^s = V_{ps} + X_{pos} \quad (1)$$

$$\bar{V}_l^s = \text{MHA}(\text{LN}(V_0^s)) + V_{l-1}^s \quad (2)$$

$$V_l^s = \text{FFN}(\text{LN}(\bar{V}_l^s)) + \bar{V}_l^s \quad (3)$$

where $X_{pos} \in \mathbb{R}^{N \times d_v}$ refers to the embedded layer at the corresponding position, which V_l^s is the hidden state of the l layer in the visual encoder. The grid feature $V^s = \{v_1^s, v_2^s, \dots, v_N^s\}$, $v_i^s \in \mathbb{R}^{d_v}$ of the image is obtained by utilizing the hidden state of the last layer in Swin-T.

3.1.2. Textual feature encoding via BERT

When performing image captioning tasks, the focus is initially on local regions of the image, gradually merging local information to generate the final description of the entire image. Similarly, we generate similar textual descriptions from image subregions, retrieving the top K most relevant descriptions from the description database of each image subregion. We define the set of retrieved textual descriptions as $T = \{t_1, t_2, \dots, t_K\}$. BERT (Kenton and Toutanova, 2019) is a popular pre-trained language model built on top of Transformer architecture, trained through unsupervised learning on large corpora. Consequently, we use the BERT model to encode similar textual descriptions T . Applying BERT to T results in the encoded textual feature $T^b = \{t_1^b, t_2^b, \dots, t_K^b\}$, where $t_i^b \in \mathbb{R}^{d_b}$ is the output of the last hidden state of the text embedding.

3.1.3. Global feature encoding via CLIP

Pretrained on massive and diverse datasets, CLIP (Radford et al., 2021) is capable of embedding both text and image into a unified mathematical space, providing natural advantages for computing cross-modal correlations. Additionally, experiments have shown that the model exhibits robustness in handling distribution shift, making it suitable for image captioning generation. Many works (Gu et al., 2021) have demonstrated the strong capability of CLIP in generalizing to

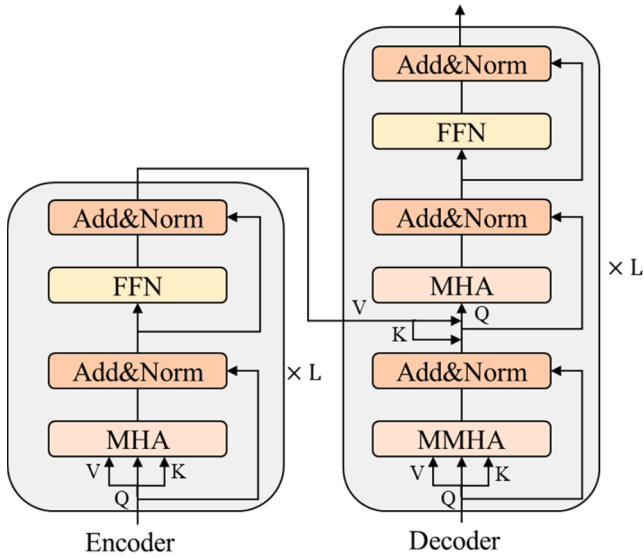


Fig. 3. The overall structure of the Transformer.

unknown domains. Therefore, we leverage CLIP's multimodal features to enrich the global correlation information of both text and visual features. Given an image I and its similar textual feature T , we encode CLIP's features as global visual $V^c \in \mathbb{R}^{d_v}$ and global textual feature $T^c \in \mathbb{R}^{d_t}$.

3.2. Transformer model

The Transformer architecture is currently a leading model in computer vision (CV) and natural language processing (NLP), featuring only self-attention mechanisms. This architecture comprises an encoder and a decoder, each consisting of 6 network layers with similar structures. The encoder consists of two sub-modules: Multi-Head Attention (MHA) and Feed-Forward Network (FFN). The decoder has a similar structure to the encoder, but includes a Mask Multi-Head Attention (MMHA) layer, which ensures information consistency during training. The overall structure of the Transformer is illustrated in Fig. 3.

The multiheaded attention layer within the Transformer initially applies a linear transformation to the input attributes, thereby generating tables of feature matrices at distinct levels. Subsequently, a scaling dot product attention is employed, merging these through the feature matrix into multiple attention heads. Lastly, employing linear mappings, these heads are interfaced and transformed into the final output representation. The mathematical formula for this process is as follows:

$$head_i = \text{Attention}\left(QW_i^Q, KW_j^K, VW_i^V\right) \quad (4)$$

$$\text{MultiHead}(Q, K, V) = \text{concat}(head_1, head_2, \dots, head_n)W^o \quad (5)$$

where $\text{Attention}(\bullet)$ represents the scaled dot-product attention mechanism, while $\text{concat}(\bullet)$ signifies the concatenation operation.

In Transformer, the network structure calculation formula for multi-head attention module and scaled dot-product attention system is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

where d_k represents the dimensionality of the key vector K .

Next, the output of the multi-head attention layer is treated as the input signal of a feedforward neural network to obtain the final output representation of the Transformer encoder. The feedforward neural network consists of two linear mapping layers and a ReLU activation

layer, where x represents the input of the feedforward neural network. The specific calculation formula for x is as follows:

$$\text{FFN}(x) = \text{ReLU}(0, xW_1 + b_1)W_2 + b_2 \quad (7)$$

where $W_1 \in \mathbb{R}^{d \times d_m}$ and $W_2 \in \mathbb{R}^{d_m \times d}$

3.3. Adaptive interaction encoder

Due to the inherent differences in expression and processing methods between text and visual modes, direct information exchange between the two is not possible. To effectively integrate visual contextual and similar text features, the Interactive Fusion Module (IFM) is employed within AIE.

We first employ the IFM module to explore the complementarity of visual and text features, which can effectively integrate different types of features. For visual information, the IFM module utilizes the Swin-T visual processing method to extract visual features, while for text information, the BERT method is used. For global information, CLIP method is selected to extract global features. The modeling is shown below:

$$P_a = \text{Add\&Norm}(\text{IFM}(T^c, V^s)) \quad (8)$$

$$P_b = \text{Add\&Norm}(\text{MHA}(T^b, T^b, T^b)) \quad (9)$$

$$\text{Add\&Norm}(f(x)) = \text{Dropout}(x + f(\text{LayerNorm}(x))) \quad (10)$$

where $\text{Add\&Norm}$ denotes the operations of residual connection and layer normalization. P_a and P_b are respectively represent the multimodal semantic information transformed via the IFM and MHA. $f(\cdot)$ is non-linear sigmoid function. The details of IFM will be elaborated in section 3.3.1.

Then, we utilized the FFN module after the IFM module:

$$H_a = \text{Add\&Norm}(\text{FFN}(P_a)) \quad (11)$$

$$H_b = \text{Add\&Norm}(\text{FFN}(P_b)) \quad (12)$$

where H_a and H_b are respectively the converted multimodal inconsistent semantics and text-level semantics in the common latent space.

Multimodal information is usually obtained from different feature extractors, and there may be redundant information among these sources, with the same or similar information repeated in multiple modalities. Noise interference, on the other hand, disturbs the real information of modality data, making it difficult to accurately model and utilize the correlation between modalities, which affects system performance and stability. Therefore, it is imperative to employ appropriate strategies to enhance the integration and interoperability of multimodal transport. We propose the Correlation Aware Module (CAM) to solve the above problems and adopt the FFN module after CAM.

$$K = [H_a; H_b] \quad (13)$$

$$I_g = \text{Add\&Norm}(\text{CAM}(K, K, K)) \quad (14)$$

$$H_g = \text{Add\&Norm}(\text{FFN}(I_g)) \quad (15)$$

where K denotes the concatenation of H_a and H_b , while I_g represents learning the rich semantics within both visual and text contexts after undergoing CAM transformation. The details of CAM will be described in section 3.3.2. After multiple layers of encoding, the interactive fusion features are fed into the decoder.

3.3.1. Interactive fusion module

Traditional parameterized projection matrix approaches may require a large number of parameters, and the scale and computational cost of the parameters need to be considered in the design and optimization

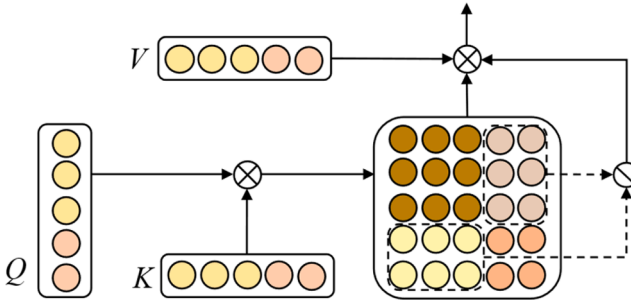


Fig. 4. The CAM architecture.

process. To reduce the complexity and computational cost of parameterized projection matrices, rule-based fusion methods, such as element-wise summation or multiplication, can be used. Specifically, for each modality feature vector, it is copied into multiple rows to form an extended matrix. These interaction operations can be element-wise operations. Apply the defined interaction operations to the feature vector extended matrix. After layer encoding, the interaction fusion features are fed to the decoder layer.

As shown in Fig. 2, the global textual feature T^c and the visual grid feature $V^s = \{v_1^s, v_2^s, \dots, v_N^s\}$ are given as two feature vectors for the visual and textual aspects.

$$V = V^s W_1^T \quad (16)$$

$$T = T^c W_2^T \quad (17)$$

To reduce computation, the text is projected onto a lower-dimensional space through the aforementioned linear mapping. Then, the global text information is transformed into an extended matrix through the following operation, which requires each row vector to be a cyclic left shift of the previous row vector.

$$T_a = \text{Aug}(T) \quad (18)$$

$$P_a = \text{Add\&Norm}(MHA(V, T_a, T_a)) \quad (19)$$

where $\text{Aug}(\cdot)$ denotes the expansion matrix. The advantage of this method lies in exploring all possible interactions between two different modal vectors, as each row of the expansion matrix shifts an element.

3.3.2. Correlation Aware module

Inspired by paper (He et al., 2021), a relevance-aware module is proposed to improve the effectiveness of multimodal fusion by explicitly modeling the inter-modality correlations. A normalized gating factor is introduced in the self-attention fusion operation to enhance the correlation and fusion quality of features across modalities, as shown in Fig. 4. The normalized gating factor plays a crucial role in dynamically reweighting and adjusting the features of different modalities to enhance the representation of relevant features. In each attention head of the module, the query and key can be formalized as:

$$\begin{aligned} Q &= \begin{pmatrix} P_a \\ H_b \end{pmatrix} W^Q = \begin{pmatrix} Q_a \\ Q_b \end{pmatrix}, \\ K &= \begin{pmatrix} P_a \\ H_b \end{pmatrix} W^K = \begin{pmatrix} K_a \\ K_b \end{pmatrix}, \\ V &= \begin{pmatrix} P_a \\ H_b \end{pmatrix} W^V = \begin{pmatrix} V_a \\ V_b \end{pmatrix} \end{aligned} \quad (20)$$

where W^Q, W^K, W^V represents the project matrix. The attention probability matrix I can be defined using four sub-matrices.

$$I = QK^\top = \begin{pmatrix} Q_a \\ Q_b \end{pmatrix} (K_a^\top, K_b^\top) = \begin{pmatrix} Q_a K_a^\top & Q_a K_b^\top \\ Q_b K_a^\top & Q_b K_b^\top \end{pmatrix} = \begin{pmatrix} I_{aa} & I_{ab} \\ I_{ba} & I_{bb} \end{pmatrix} \quad (21)$$

where I_{aa} and I_{bb} represent within-modality attention, while I_{ab} and I_{ba} represent between-modality attention. The gate G is used to adaptively adjust the correlation between the two modalities.

$$I_g = G \odot I = \begin{pmatrix} 1 & g_{ab} \\ 1 - g_{ab} & 1 \end{pmatrix} \odot \begin{pmatrix} I_{aa} & I_{ab} \\ I_{ba} & I_{bb} \end{pmatrix} = \begin{pmatrix} I_{aa} & g_{ab} I_{ab} \\ (1 - g_{ab}) I_{ba} & I_{bb} \end{pmatrix} \quad (22)$$

$$g_{ab} = \frac{\sum_i \exp(Q_a K_a^\top)_i}{\sum_i \exp(Q_a K_a^\top)_i + \sum_j \exp(Q_a K_b^\top)_j} \quad (23)$$

where $G \in \mathbb{R}^{2 \times 2}$ is the adaptive correlation gate, representing a scalar that is the sum of normalized attention weights on text key and value vectors.

3.4. Adaptive Interactive decoder

To begin with, we represent the vector of the word $w_t \in \mathbb{R}^d$ as a combination of word embedding and positional encoding for the t -th word. Therefore, the input matrix for time step t is $W_{t-1} = \{w_1, w_2, \dots, w_{t-1}\} \in \mathbb{R}^{d \times t}$. We denote the embedded existing caption generated by the classic captioning model as $E = \{e_1, e_2, \dots, e_k\} \in \mathbb{R}^{d \times k}$, where k is the length of existing caption.

$$R = \text{Add\&Norm}(MMHA(E, E, E)) \quad (24)$$

$$C = \frac{1}{k} \sum_{i=1}^k r_i \quad (25)$$

where $R = \{r_1, r_2, \dots, r_k\} \in \mathbb{R}^{d \times k}$ denotes the semantic information derived from word embeddings. $C \in \mathbb{R}^{d \times 1}$ is the context vector which can be viewed as sentence encoding of the existing caption.

The decoder in the generation process of step t , receives inputs b and context vector C . Subsequently, these are fed into our proposed Adaptive Guidance Module (AGM) module for modelling the relationships between sequences, while concurrently adaptively capturing global information. Ultimately, the output of our AGM (denote as F) is funneled into a feed-forward neural network (FFN).

$$F = \text{Add\&Norm}(AGM(C, H_g, V^c, T^c)) \quad (26)$$

$$F_h = \text{Add\&Norm}(FFN(F)) \quad (27)$$

We augment the computation of self-attention within the decoder by incorporating the context vector C , into the key and value matrices. In this scenario, the query vector for each time step captures not only information regarding sub sequences but also adaptively modulates multimodal features at each instant through weighted similarity to text global T^c and visual global V^c . The calculation of AGM as below:

$$S = \frac{C \cdot (V^c)^T}{\|C\| \|V^c\|} + \frac{C \cdot (T^c)^T}{\|C\| \|T^c\|} \quad (28)$$

$$F = MHA(S \cdot \Phi(H_g)) \quad (29)$$

Where $\Phi(\cdot)$ is a projection head. S denotes the coefficient utilizing the cosine similarity of CLIP to formulate the weighted multimodal feature.

After L identical decoding layers, the output F_h^L is provided to the classifier layer in the vocabulary table to predict the subsequent word. We designate the predicted sentence $Y_{t-1} = \{y_0, y_1, \dots, y_{t-1}\}$ from the previous time step $t-1$. Then, the conditional probability distribution for the current time t word is $p(y_t | Y_{t-1})$, which can be calculated by:

$$p(y_t | Y_{t-1}) = \text{softmax}(W_y F_h^L) \quad (30)$$

where $W_y \in \mathbb{R}^{|V| \times d}$, and $|V|$ is the number of words in the vocabulary.

3.5. Training and objectives

Given the target ground truth sequence $y_{1:T}^*$ and the captioning model with parameter θ , DAIT is trained by minimizing cross entropy (XE) L_{XE} :

$$L_{XE}(\theta) = - \sum_{t=1}^T \log(p_{\theta}(y_t^* | y_{1:t-1}^*)) \quad (31)$$

Since reinforcement learning has been used to train captioning models, we follow this training strategy to optimize non-differentiable metrics, and then seek the minimum negative expected score from the initialization of the trained model under cross-entropy:

$$L_R(\theta) = -E_{y_{1:T} \sim p_{\theta}}[r(y_{1:T})] \quad (32)$$

where r is the CIDEr-D score function. We directly optimize the non-differentiable metrics with Self-Critical Sequence Training (SCST) (Rennie et al., 2017), and the gradient can be approximated:

$$\nabla_{\theta} L_R(\theta) \approx - (r(y_{1:T}^s) - r(\hat{y}_{1:T})) \nabla_{\theta} \log p_{\theta}(y_{1:T}^s) \quad (33)$$

where $y_{1:T}^s$ denotes a result sampled from probability distribution and $r(\hat{y}_{1:T})$ is the baseline score of greedily decoding.

4. Experimental setup

In this section, we initially introduced two datasets based on the multimodal image description generation task. Subsequently, we delineated how the model's overall experimental evaluation metrics scores were determined. Lastly, we detailed the precise experimental setup for the model.

4.1. Dataset

The MS COCO dataset is widely utilized in the field of image captioning and consists of a total of 164,062 images. This consists of 82,783 images designated for training, 40,504 for validation, and 40,775 for testing. Each image in the dataset was annotated by multiple Amazon Mechanical Turk (AMT) personnel with a minimum of five captions. A popular approach in previous research, the Karpathy splits method (Karpathy & Fei-Fei, 2015) was used to evaluate the dataset. The dataset was divided into subsets of 123,287 images, with 113,287 for training, 5,000 for validation, and 5,000 for testing. The online evaluation is performed on the MS COCO test server and is not accessible to the general public. For the Flickr30K dataset (Young et al., 2014), 29014, 1000 and 1000 images are used to train, validate, and test respectively.

4.2. Evaluation metrics

To evaluate the quality of generated captions, researchers have proposed a range of automatic metrics based on the similarity between generated and ground truth captions. In accordance with the standards set forth in the 2015 MS COCO Image Captioning Challenge, the results are reported utilizing five metrics. These include BLEU (Papineni et al., 2002), employed to gauge the degree of co-occurrence of N-tuples generated by the model with those in the reference translation during machine translation. METEOR (Denkowski et al., 2014), utilizing WordNet for the calculation of affixes, definitions, etc., while considering synonyms. ROUGE (Lin et al., 2004), which calculates precision and recall measures from the concurring vector of the longest common sub-sentences. CIDEr (Vedantam et al., 2015), which estimates the concurrency between an image description's score calculated based on the weight of each n-tuple in the text description, and the image tag's information. SPICE (Anderson et al., 2016), which determines the F-score of the objects, attributes, and correspondences between the true

Table 1

Performance comparisons on MS-COCO Karpathy test split under the optimization stage. All values are reported as percentage (%).

Method	B-1	B-4	M	R	C	S
MA	81.2	39.5	29.4	59.3	132.0	23.0
DLCT	81.4	39.8	29.5	59.1	133.8	23.0
PureT	82.1	40.9	30.2	60.1	138.2	24.2
CaMEL	82.8	41.3	30.2	60.1	140.6	23.9
CTX + M2	81.5	39.7	30.0	59.5	135.9	23.7
VinVLB	–	40.9	30.9	–	140.4	25.1
HAAV	–	41.0	30.2	–	141.5	23.9
DAIT(Ours)	83.3	41.6	31.2	60.9	143.6	24.9

and predicted image descriptions through the quantification of semantic information.

4.3. Implementation details

The proposed model retrieves the top 4 most relevant text descriptions from a description database when given a query for an image subregion, enabling relevant information to be obtained for generating captions. The model architecture is illustrated in Fig. 3, featuring an encoder that comprises frozen Swin-T, BERT, and CLIP parameters, alongside a decoder with 3 layers. The model employs self-attention mechanisms with 8 heads and a dimension of 512, with dropout regularization applied after each attention and feed-forward layer using a keep probability of 0.9. A learning rate scheduling strategy with a warmup period of 20,000 iterations is followed during the cross-entropy (XE) pre-training stage. After the XE pre-training stage, the model is fine-tuned using the CIDEr reward and is trained for 30 epochs with a learning rate of 5×10^{-6} . For parameter optimization, the Adam optimizer (Kingma and Adam, 2014) with betas of (0.9, 0.999) is utilized, with the models trained with a batch size of 50 and implemented using the PyTorch framework on a single Nvidia 3090 GPU.

5. Experimental results and analysis

In this section, we analyze the experimental outcomes between our model and multiple contemporary baselines models. Furthermore, we thoroughly delve into the efficacy and advantages of each component of our model.

5.1. Captioning performance

Table 1 presents the performance results of the model on the offline “Karpathy” dataset. When comparing performance, this study selected seven comparative models, all of which utilized a generator method based on the encoder-decoder framework. The compared models include: MA (Fei, 2021), DLCT (Luo et al., 2021), PureT (Wang et al., 2022), CaMEL (Barraco et al., 2022), CTX + M2 (Kuo and Kira, 2022), VinVLB (Ramos et al., 2023), HAAV (Kuo and Kira, 2023). B1, B4, M, R, C, S respectively represent BELU-1, BELU-4, METEOR, ROUGE-L, CIDEr, SPICE. Bold data shows the highest position under each index.

We compared models solely utilizing visual information as input (MA, DLCT, CaMEL). These three models exclusively fuse image information without external textual information, yielding accuracy rates lower than other models incorporating external knowledge on the MS-COCO dataset. Our model is based on images and similar text as input, enhanced through external text context knowledge, accomplishing an average enhancement of 4.2% over single vision models' accuracy rates. This signifies that text and visual patterns aid image captioning, especially text featuring more identical characteristics which is conducive to optimizing image captioning.

Secondly, we compared models entailing both visual and text context information as input (PureT, CTX + M2, VinVLB, HAAV). These four models all incorporate external textual semantic knowledge as part of

Table 2

Performance comparisons with other advanced single models on the official MS COCO test server. All values are reported as percentage (%).

	B-1		B-2		B-3		B-4		M		R		C	
	c5	c40	c5	c40	c5	c40	c5	c40	c5	c40	c5	c40	c5	c40
DLCT	82.0	96.2	66.9	91.0	52.3	83.0	40.2	73.2	29.5	39.1	59.4	74.8	131.0	133.4
MA	81.7	96.2	66.5	90.9	51.8	82.7	39.7	72.5	29.3	38.7	59.2	74.2	130.1	132.4
DRT	82.7	96.5	67.7	91.5	53.1	83.4	40.9	73.6	29.6	39.0	59.8	75.0	132.2	133.9
PureT	82.8	96.5	68.1	91.8	53.6	83.9	41.4	74.1	30.1	39.9	60.4	75.9	136.0	138.3
CaMEL	83.2	97.3	68.3	92.7	53.6	84.8	41.2	74.9	30.2	39.7	60.2	75.6	137.5	140.0
DAIT(Ours)	83.6	98.1	68.0	93.5	54.4	85.2	41.2	74.1	31.1	40.6	60.2	76.2	138.3	141.2

Table 3

Performance comparisons on Flickr30K test split under the optimization stage. All values are reported as percentage (%).

Method	B-1	B-4	M	R	C	S
CaMEL	72.3	33.4	23.8	52.3	77.4	17.1
CTX + M2	72.8	34.6	24.5	52.6	77.7	17.5
VinVLB	73.2	35.3	25.0	53.1	78.1	17.8
OFA	74.7	35.8	25.2	52.8	77.2	18.3
BLIP-2	76.0	36.1	25.8	53.6	79.2	19.1
HAIV	75.0	35.8	25.3	53.4	78.8	18.6
DAIT(Ours)	75.7	36.4	26.3	54.3	80.1	19.5

the model's input, but their mode of fusion involves straightforward connections between different feature, which can result in the loss of details and a reliance on fusion features, disregarding irrelevant textual information. Our proposed DAIT addresses the issues prevalent in these models, achieving an average improvement in accuracy rate of 2.3 % compared to the aforementioned four models. This suggests that modeling consistency around multimodal features can significantly enhance the efficacy of image captioning.

Table 2 presents the comparison results of the proposed DAIT model and other established models on the MS COCO test server. A fair comparison was conducted between a single model and state-of-the-art models, providing two evaluation settings: five caption references per image (c5) and 40 caption references per image (c40). Based on the results in the table, the single model exhibits superior performance over other models in multiple metrics. Specifically, it outperforms other models in BLEU-1 (c5 and c40), BLEU-2 (c40), BLEU-3 (c5 and c40), METEOR (c5 and c40), ROUGE (c40), and CIDEr (c5 and c40). Additionally, it achieves competitive performance in BLEU-1 (c5), ROUGE (c5), and BLEU-4 (c5 and c40). These results demonstrate that the DAIT model achieves satisfactory outcomes in the image captioning task and surpasses other established models in multiple evaluation metrics.

As illustrated in Table 3, we present the experimental results of DAIT on the Flickr30K (Young et al., 2014) dataset, benchmarking several state-of-the-art baselines based on transformers. CaMEL (Wang et al., 2022) proposes harnessing two interdependent language models to mutually learn during the training process. CTX + M2 (Kuo and Kira, 2022) perform inference on assumed causal graphical models and adding an auxiliary input to represent missing information. VinVLB (Ramos et al., 2023) proposes to jointly process images and retrieved captions using pre-trained V&L BERT in the encoder. OFA (Wang et al., 2022) propose a task-agnostic and modality agnostic framework. BLIP-2 (Li et al., 2023) proposes a generic and efficient pre-training strategy that bootstraps vision-language pre-training from off-the-shelf frozen pre-trained models.

The results indicates that our method surpasses all Transformer-based methods in terms of BLEU-4, ROUGE-L, CIDEr and SPICE when compared with the SOTA approach, as well as exhibiting competitive performance on BLEU-4 and METEOR. In particular, it advances the current state-of-the-art on ROUGE-L and CIDEr by 1.7 and 2.4 points, respectively. Simultaneously, it also indicates that the performance improvement produced by our proposed DAIT model is equally effective on different datasets.

5.2. Qualitative analysis

Fig. 5 presents illustrative examples and a cross-comparison with the CTX + M2 model, which shares similarities with our work and achieves optimal performance. Additionally, a contrast is drawn between the generated descriptions and the manually catalogued text (ground truth, GT) provided in the data set. The results demonstrate that the DAIT model is proficient in extracting efficacious information from both textual and visual cues, while utilizing multi-grained contextual semantic knowledge to produce accurate description sentences. In the first and fourth samples, the baseline model neglects to include information regarding small objects and positional contexts in the image, whereas the descriptions generated by the DAIT model encompass these details with precision. For instance, the sentences generated by the DAIT model, "is cooking in a pot" and "a skateboard and bike," accurately describe all objects and their spatial relationships within the image. Furthermore, in the second and third samples, the description sentences generated by the DAIT model, "A man in a brown shirt" and "next to a homeless man," are more vivid and specific than those generated by the CTX + M2 model, providing an in-depth description of the contextual information of the targets in the image. Overall, these examples demonstrate the superior performance of the DAIT model over the baseline model in the generation of image descriptions.

5.3. Ablation studies

Firstly, the performance of DAIT at different depths was studied, as shown in Fig. 6. It can be observed from the graph that the performance scores of the encoder and decoder gradually improve with increasing depth. This indicates that increasing depth helps to enhance the model's learning and expression abilities, enabling it to better capture semantics and image features when processing text and visual information. However, the model's performance begins to decline from the third layer onwards. This may be due to overfitting of the training data on deeper layers, where the model becomes too focused on details and local features, resulting in a decrease in overall semantic understanding and performance on image information. Additionally, it is worth noting that the decoder maintains the context vector from the second layer, and consistently achieves higher scores than other layers. The encoder achieves its best score at the third layer. This suggests that a shallow decoder can better understand contextual information and generate more accurate results, while the third layer of the encoder contributes the best feature representation.

To demonstrate the effectiveness of all proposed modules in DAIT, we conducted ablation experiments as shown in Table 4. The first row shows the scores obtained when using visual and textual Transformer as the baseline model, which serves as the initial performance of the model. The second row shows the scores obtained when only the AID module is selected, and the results demonstrate that AID enhances the generation of descriptions by providing adaptive contextual information. Furthermore, we investigated the impact of using the IFM and CAM modules separately on the description model based on the AID module, such as the third and fourth rows. The IFM module increased the CIDEr score from 137.8 to 139.9, while the CAM module increased the CIDEr score

Image	Captions
	Ours: Some food is cooking in a pot in the kitchen. CTX+M2: Some food is on the kitchen table. GT1: A commercial stainless kitchen with a pot of food cooking. GT2: Some food sits in a pot in a kitchen. GT3: Food cooks in a pot on a stove in a kitchen.
	Ours: A man in a brown shirt riding an elephant in a river. CTX+M2: A man riding an elephant in a river. GT1: man riding an elephant into water surrounded by forest. GT2: A man in a brown shirt rides an elephant into the water. GT3: A man riding an elephant into some water of a creek.
	Ours: a woman is walking down the street next to a homeless man. CTX+M2: a person with a shopping cart on a street. GT1: People walking past a homeless man begging on a city street. GT2: a homeless man holding a cup and standing next to a shopping cart on a street. GT3: People are walking on the street by a homeless person.
	Ours: A person on a skateboard and bike at a skate park. CTX+M2: A person on a skateboard at a park. GT1: A skateboarder jumps into the air as he performs a skateboard trick. GT2: Athletes performing tricks on a BMX bicycle and a skateboard. GT3: A man on a skateboard performs a trick at the skate park.

Fig. 5. Qualitative Comparison of Experimental Results.

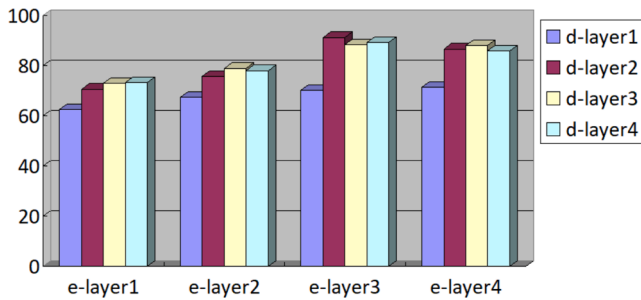


Fig. 6. Visualizing the Score of Encoder-Decoder Layer. The X-axis represents the encoder layer from 1st to 4th, while the Y-axis represents the decoder layer from 1st to 4th. All reported values are from the offline Karpathy test set.

from 137.8 to 140.4. When using the IFM and CAM modules simultaneously fused as AIE (i.e., IFM + CAM) as shown in the fifth row, the best performance of the encoder was achieved with a CIDEr score improvement of 3.1 points. This further confirms that mining the complementarity between modalities and reducing interference can improve the

Table 4

Ablation on different variants of the DAIT.

Encoder	Decoder	B-1	B-4	M	R	C
–	–	81.2	39.6	29.5	59.2	135.7
–	AID	81.7	39.9	30.1	59.5	137.8
w/IFM	AID	81.9	40.2	29.0	60.5	139.9
w/CAM	AID	82.1	40.6	30.8	60.2	140.4
AIE	–	82.2	40.1	30.5	60.1	138.8
AIE	AID	83.3	41.6	31.2	60.9	143.6

model's performance. Finally, the combination of the AIE and AID modules resulted in the best score, further validating the effectiveness of these modules.

5.4. Visualization analysis

We utilize text from manually labeled data to assist in image captioning tasks. In image captioning, natural language descriptions are typically generated for images. However, since the semantic connection between the image and its description is not direct, using text from

Image				
Text	a brick building a sport motorbike on the street close to a building	a bunch of cars an image of a cars a busy intersection a very busy street	at a skate park on a bicycle A man on skateboard jumps into the air	some horses an old building horses eating grass next to a church
Model Captioning	Baseline: a blue motorcycle is parked on the street. DAIT: a sport motorbike parked in front of a red brick building.	Baseline: a group of cars driving on the road. DAIT: a bunch of cars driving on a very busy highway.	Baseline: a person on a skateboard at a park. DAIT: a man falls off his skateboard in a skate park.	Baseline: some horses are grazing in a field. DAIT: three horses are grazing in a field in front of an old building.

Fig. 7. Description generated by visualizing text retrieval.

manually labeled data can serve as auxiliary information to aid in generating accurate and coherent descriptions. In this paper, we use a model without text assistance as the baseline model in our experiments and compare it to a model that uses text assistance. As shown in Fig. 7, the baseline model without text assistance has a poorer performance and cannot fully utilize the finer details in the images. Our experimental results demonstrate that the DAIT method has a significant improvement in the generated descriptions compared to the baseline model, indicating that incorporating text assistance can enhance the image captioning task. The DAIT method is able to better utilize text information to complement the image content, thereby improving the quality and accuracy of the generated descriptions. Moreover, the DAIT method is also able to reduce noise interference from the text, thereby enhancing the effectiveness of description generation.

6. Conclusions

In this paper, we propose a model named Dual-Adaptive Interactive Transformer (DAIT), which aims to effectively integrate textual and visual contextual information to understand image content. During encoding, the Adaptive Interactive Encoder (AIE) expands and fuses visual and textual feature vectors to more fully utilize the complementary nature of the two modalities. Additionally, through the introduction of a normalization gating factor, the AIE module effectively adjusts and optimizes the interaction between the two modalities, reducing potential noise. During decoding, the Adaptive Interactive Decoder (AID) uses a weighted multi-modal feature based on the similarity between the textual and visual branches. Extensive experiments were conducted on the MS COCO image captioning dataset to verify the model’s performance, and it demonstrated excellent performance surpassing other methods. This proves the effectiveness and superiority of the DAIT model in image captioning tasks.

Our model focuses on the inconsistency in multi-modal content’s semantics. Nonetheless, due to limited or inadequate semantics of images and text, there remains a challenge in accurately detailing the statements. Moving forward, we are committed to incorporating external knowledge or a multi-modal knowledge graph to augment common meaning and contextual knowledge in image description, thereby enhancing the ability of our model to generate content.

CRediT authorship contribution statement

Lizhi Chen: Conceptualization, Investigation, Formal analysis, Writing – original draft. **Kesen Li:** Data curation, Methodology, Software, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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